

5058 Homework1

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1. Bayes' Theorem

Definitions: T_0, T_1 : transmitted dot and dash. R_0, R_1 : received dot and dash.

Given:

- $P(R_1|T_0) = 2/5 \implies P(R_0|T_0) = 3/5$
- $P(R_0|T_1) = 1/3 \implies P(R_1|T_1) = 2/3$
- $P(T_0) = 5/8, \quad P(T_1) = 3/8$

(a)

Total probability of R_0 :

$$P(R_0) = P(R_0|T_0)P(T_0) + P(R_0|T_1)P(T_1) = \frac{3}{5} \cdot \frac{5}{8} + \frac{1}{3} \cdot \frac{3}{8} = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$$

Hence, $P(R_1) = 1 - P(R_0) = 1/2$.

(i) Received signal is a dot:

$$P(T_0|R_0) = \frac{P(R_0|T_0)P(T_0)}{P(R_0)} = \frac{(3/5)(5/8)}{1/2} = \frac{3}{4} = 0.75$$

(ii) Received signal is a dash:

$$P(T_1|R_1) = \frac{P(R_1|T_1)P(T_1)}{P(R_1)} = \frac{(2/3)(3/8)}{1/2} = \frac{1}{2} = 0.5$$

(b)

Given independence, $P(T_i T_j | R_0 R_0) = P(T_i | R_0) P(T_j | R_0)$. From (a), $P(T_0 | R_0) = 3/4$ and $P(T_1 | R_0) = 1/4$.

The distribution for sent messages (T_i, T_j) given $R_0 R_0$:

- $P(T_0 T_0 | R_0 R_0) = \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16} = 0.5625$
- $P(T_0 T_1 | R_0 R_0) = \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16} = 0.1875$
- $P(T_1 T_0 | R_0 R_0) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16} = 0.1875$
- $P(T_1 T_1 | R_0 R_0) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16} = 0.0625$

2. Characteristic Function

Given the PDF:

$$f(x) = \frac{1}{2} e^{-|x|}, \quad x \in \mathbb{R}$$

(a) Characteristic function $\phi_X(t)$

$$\begin{aligned} \phi_X(t) &= E[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} \frac{1}{2} e^{-|x|} dx \\ &= \frac{1}{2} \int_{-\infty}^0 e^{(1+it)x} dx + \frac{1}{2} \int_0^{\infty} e^{-(1-it)x} dx \\ &= \frac{1}{2} \left[\frac{e^{(1+it)x}}{1+it} \right]_{-\infty}^0 + \frac{1}{2} \left[\frac{e^{-(1-it)x}}{-(1-it)} \right]_0^{\infty} \\ &= \frac{1}{2} \left(\frac{1}{1+it} + \frac{1}{1-it} \right) \\ &= \frac{1}{2} \left(\frac{1-it+1+it}{(1+it)(1-it)} \right) = \frac{1}{1+t^2} \end{aligned}$$

(b) Mean and Variance

Using the property $E[X^n] = \frac{1}{i^n} \phi_X^{(n)}(0)$:

First and second derivatives of $\phi_X(t) = (1 + t^2)^{-1}$:

$$\begin{aligned}\phi'_X(t) &= -(1 + t^2)^{-2} \cdot 2t = \frac{-2t}{(1 + t^2)^2} \\ \phi''_X(t) &= \frac{-2(1 + t^2)^2 - (-2t) \cdot 2(1 + t^2) \cdot 2t}{(1 + t^2)^4} = \frac{6t^2 - 2}{(1 + t^2)^3}\end{aligned}$$

Mean:

$$E[X] = \frac{1}{i} \phi'_X(0) = \frac{1}{i}(0) = 0$$

Variance:

$$\begin{aligned}E[X^2] &= \frac{1}{i^2} \phi''_X(0) = -1 \cdot (-2) = 2 \\ \text{Var}(X) &= E[X^2] - (E[X])^2 = 2 - 0^2 = 2\end{aligned}$$

3. Autocorrelation

Let $x = X(t)$ and $y = X(t + \tau)$. Since $X(t)$ is stationary Gaussian with $E[X] = 0$:

- $E[x] = E[y] = 0$
- $\text{Var}(x) = \text{Var}(y) = \sigma^2$
- $E[xy] = R_X(\tau) = \sigma^2 \rho(\tau)$

The term $\frac{xy}{|xy|}$ is the sign function $\text{sgn}(xy)$. Thus:

$$Z(t) = \frac{1}{2}[1 + \text{sgn}(xy)] = \begin{cases} 1 & \text{if } xy > 0 \\ 0 & \text{if } xy < 0 \end{cases}$$

Assuming ergodicity, the time average $\bar{Z}(\tau) = E[Z(t)]$:

$$E[Z] = P(xy > 0) = P(x > 0, y > 0) + P(x < 0, y < 0)$$

For bivariate normal variables with correlation ρ , the orthant probability is:

$$P(x > 0, y > 0) = P(x < 0, y < 0) = \frac{1}{4} + \frac{1}{2\pi} \arcsin(\rho)$$

Summing these gives:

$$\bar{Z}(\tau) = 2 \left(\frac{1}{4} + \frac{1}{2\pi} \arcsin(\rho) \right) = \frac{1}{2} + \frac{1}{\pi} \arcsin(\rho)$$

Using the trigonometric identities $\arcsin(\rho) = \frac{\pi}{2} - \arccos(\rho)$ and $\arccos(-\rho) = \pi - \arccos(\rho)$:

$$\begin{aligned} \bar{Z}(\tau) &= \frac{1}{2} + \frac{1}{\pi} \left(\frac{\pi}{2} - \arccos(\rho) \right) \\ &= 1 - \frac{1}{\pi} \arccos(\rho) \\ &= \frac{1}{\pi} [\pi - \arccos(\rho)] \\ &= \frac{1}{\pi} \arccos[-\rho(\tau)] \end{aligned}$$

4. Radar System

Given: $W \sim N(0, \sigma^2)$ with $\sigma^2 = 1/9 \implies \sigma = 1/3$. $X = \theta + W$.

(a)

$H_0 : \theta = 0$ (no aircraft)

$H_1 : \theta = 1$ (aircraft present)

(b)

Significance level $\alpha = 0.05$. Under H_0 , $X \sim N(0, 1/9)$. We reject H_0 if $X > c$.

$$P(X > c | H_0) = 0.05 \implies P\left(\frac{X - 0}{1/3} > \frac{c}{1/3}\right) = 0.05$$

From Z-table, $3c = 1.645 \implies c = 0.5483$. **Test:** Reject H_0 if $X > 0.5483$.

(c)

Type II error $\beta = P(\text{accept } H_0 | H_1)$. Under H_1 , $X \sim N(1, 1/9)$.

$$\begin{aligned}\beta &= P(X \leq 0.5483 | H_1) = P\left(\frac{X - 1}{1/3} \leq \frac{0.5483 - 1}{1/3}\right) \\ &= \Phi(-1.355) = 1 - 0.9123 = 0.0877\end{aligned}$$

(d)

Observe $X = 0.6$. Test at $\alpha = 0.01$:

$$3c = z_{0.01} = 2.326 \implies c = 0.7753$$

Since $0.6 < 0.7753$, there is **not enough evidence** to reject H_0 .

(e)

"Missing a present aircraft" is β . We want $\beta < 0.05$.

$$P(X \leq c | H_1) = 0.05 \implies \frac{c - 1}{1/3} = -1.645 \implies c = 1 - \frac{1.645}{3} = 0.4517$$

Smallest significance level α :

$$\alpha = P(X > 0.4517 | H_0) = P(Z > 3 \times 0.4517) = P(Z > 1.355) = 1 - 0.9123 = 0.0877$$

(f)

Likelihood ratio test: $\Lambda(x) = \frac{f(x|H_1)}{f(x|H_0)} > \eta$

$$\begin{aligned}\frac{\frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x-1)^2}{2\sigma^2}\right]}{\frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{x^2}{2\sigma^2}\right]} &> \eta \\ \exp\left[\frac{x^2 - (x-1)^2}{2\sigma^2}\right] &> \eta \\ \exp\left[\frac{2x-1}{2/9}\right] &> \eta \\ x &> \frac{1}{9} \ln \eta + \frac{1}{2}\end{aligned}$$